

Digital Dentistry Workflow

A Quick-Reference Cheatsheet and Checklist for Dental 3D Printing Workflows



Chapter 1: The Core Digital Workflow

To ensure maximum success and predictable results when printing clinical models, guides, and splints, you must master the fundamental digital loop.

This cheatsheet serves as your clinic's quick-reference guide. Keep it in your lab or near your 3D printing station to verify settings and standard operating procedures (SOPs).

Here is a quick look at the first two stages of the digital loop:

1. Data Capture (Scan)

Use an intraoral scanner to capture the patient's mouth. Ensure clear scan data around margins and check the occlusion alignment visually. Export files in STL or PLY format.

2. Design (CAD)

Import files into dental design software. Design the desired guide, splint, or teeth alignment model. Check boundary thicknesses (min 1.5mm for splints, 2.0mm for surgical guides) and export as solid STLs.

Chapter 2: Build Preparation & Slicing

Stage 3 of the loop is 'Slicing', where digital designs are prepared for the printer. Correct slicing settings are crucial to prevent print detachments.

Slicing Setup Guide:

1. Slicer Selection: Use the software recommended by your printer manufacturer (e.g., Chitubox, Formlabs PreForm, or Asiga Composer).
2. Layer Height: Use 50-micron (0.05mm) layers for aligner models and splints. Use 100-micron layers for study models to print faster.
3. Model Orientation: Tilt models at a 30 to 45 degree angle relative to the build platform. Printing flat causes excessive peel forces that break print layers.
4. Support Struts: Generate automatic supports with a density of 50-60%. Place heavy supports on the base and light supports on critical margins to make removal clean.

Rule of Thumb: Never place support tips directly on tooth margins or contact areas. Place them on non-critical regions like the palate or base.

Chapter 3: Post-Processing Reference

Post-processing (washing and curing) changes the printed liquid resin into a solid, safe, biocompatible dental device. Follow this strict guide:

Washing Guidelines:

- Alcohol Type: Use high-purity (99%+) Isopropyl Alcohol (IPA) or Glycol Ether (TPM).
- Wash Time: Wash in an active wash tank for 3 to 5 minutes. Over-washing weakens the resin matrix.
- Drying: Let the models completely air dry (or blow dry with compressed air) before curing. Curing wet resin causes white chalky deposits.

Post-Curing Guidelines:

- UV Curing Chamber: Always cure in a professional UV chamber with a rotating glass plate.
- Temperature: Many medical-grade resins require heat (e.g. 60 deg C) during curing to maximize strength.
- Curing Time: Cure for 5 to 15 minutes, rotating parts regularly. Fully cured parts will lose any sticky or soft feel.

Chapter 4: Troubleshooting Checklist

If your prints are failing, run through this quick troubleshooting checklist:

Common Issues & Fixes:

1. Print is separating from build plate: Re-calibrate the build plate level. Increase the 'Bottom Exposure Time' in your slicer settings to ensure the first layers stick firmly.
2. Model has warped or fits poorly: Check your washing time. If washed too long in alcohol, parts will swell. Ensure complete post-curing to stabilize the shape.
3. Cloudiness in print detail: Replace or clean the FEP film at the bottom of the resin vat. Clean the printer's optical glass cover panel.
4. Parts break during support removal: Remove supports BEFORE post-curing when the resin is still slightly flexible (green state).

For technical support or shape-memory aligner inquiries, visit align-all.com or contact Align-All Clinical Support directly.